Physical Chemistry (I)

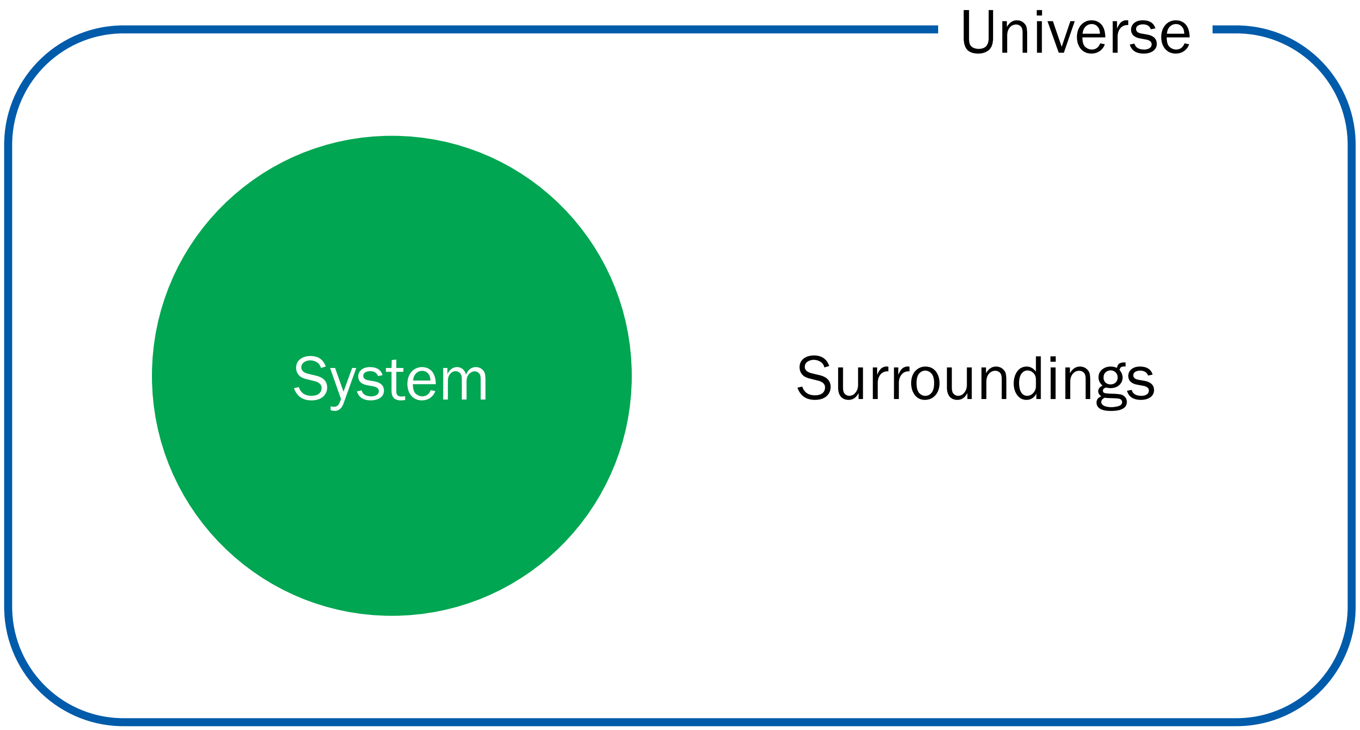
Lecture 03

Kinetic Molecular Theory

In the Last Class

- Basic Concepts
  - System: properties, equilibrium
  - System state and state change
- The Properties of Gases
  - Why gases?
  - Perfect gas

System and Surroundings



System Properties

- A system can have different physical properties
- A single value will be assigned for each property

- Useful classification
  - Extensive properties: $f(\mathrm{A}) + f(\mathrm{B}) = f(\mathrm{A} + \mathrm{B})$
  - Intensive properties: $f(\mathrm{A}) + f(\mathrm{B}) \neq f(\mathrm{A} + \mathrm{B})$

Equilibrium

- An isolated system is in equilibrium when its properties remain constant with time

- Mechanical equilibrium: constant p and V
- Diffusive equilibrium: constant n
- Thermal equilibrium: constant T

State of a System

- The state of a system is defined by its properties

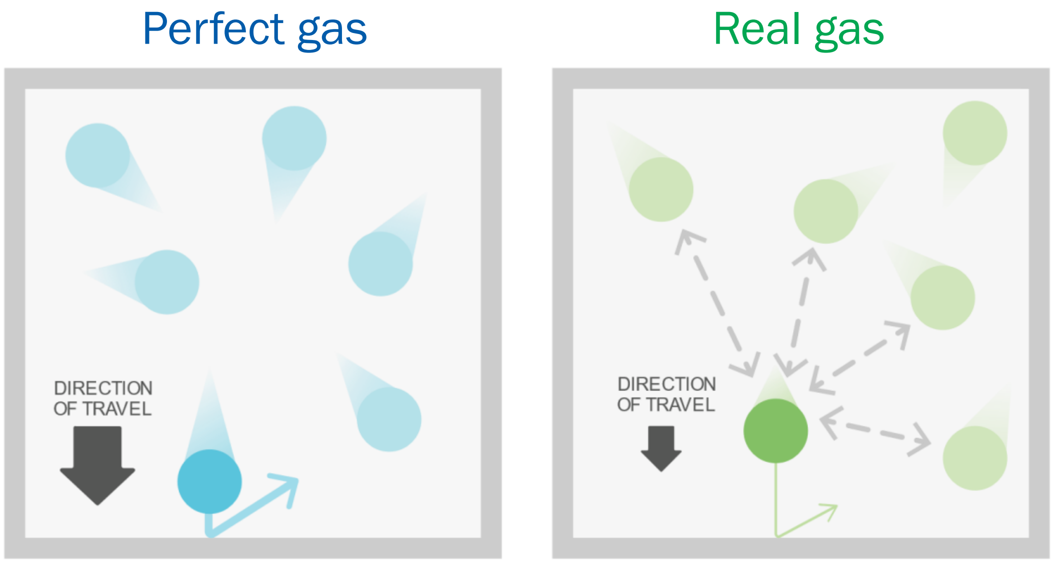
- For a pure substance: (n, V, p, T)

- State change: $(n_1, V_1, p_1, T_1) \rightarrow (n_2, V_2, p_2, T_2)$

Last Class

Perfect Gas

- Perfect gas (ideal gas): no interactions
- Real gas: finite interactions between molecules



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Equation of State

- The state of a pure gas is defined by (n, V, p, T)
- Equation of state: relationship of the 4 variables

$$p = f(T, V, n)$$

- Equation of state of a perfect gas: perfect gas law

$$p = nRT/V$$

gas constant

Last Class

Kinetic Molecular Theory

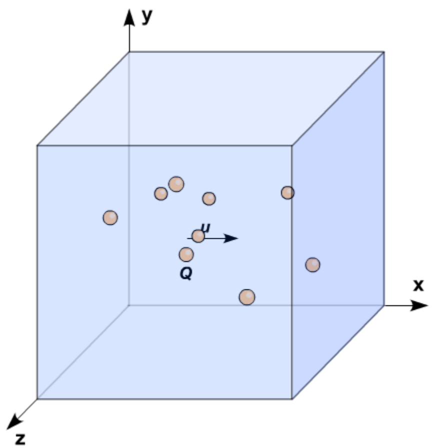
Total energy = Σ (kinetic energies of molecules)

1. Molecules with mass m in ceaseless random motion
2. Molecules with negligible size
3. Elastic collisions: the total kinetic energy is conserved

Last Class

Molecular Velocities

Each molecule has a different velocity



How can we determine the information about velocity?

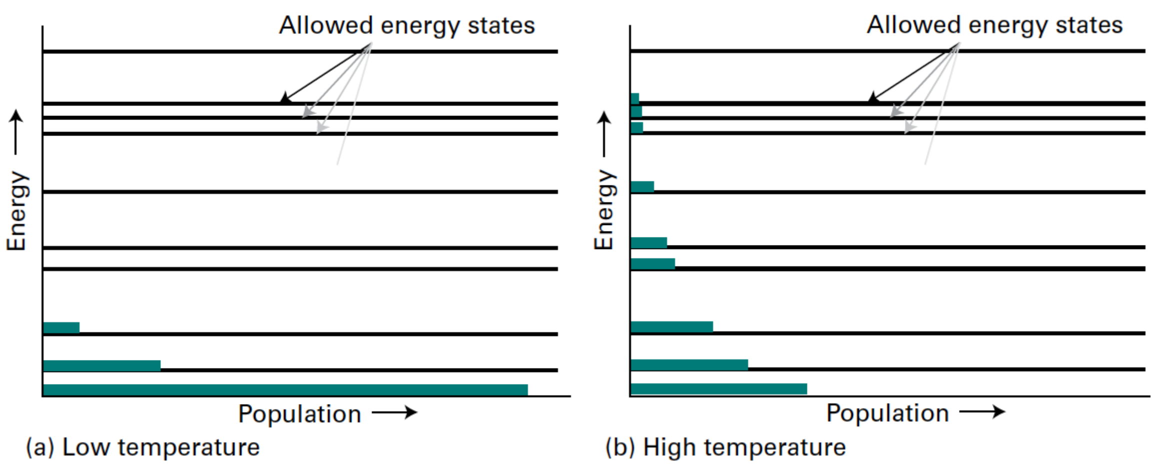
(Image: https://www.pinterest.ca/pin/314970567674652846/)

Boltzmann Distribution

Boltzmann constant

$$\text{Probability} \propto e^{-E/k_{\mathrm{B}}T}$$

(at thermal equilibrium)



(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., Prologue Figure 2)

Probability: Example

Assume that we roll a fair dice. Probability is denoted as $p(k)$.

$$p(1) = p(2) = p(3) = p(4) = p(5) = p(6) = 1/6$$

- Normalization: $p(1) + p(2) + p(3) + p(4) + p(5) + p(6) = 1$

$$\sum_{k=1}^{6} p(k) = 1$$

- Average: $\langle k \rangle = (1 + 2 + 3 + 4 + 5 + 6)/6 = 3.5$

$$\langle k \rangle = 1 \cdot p(1) + 2 \cdot p(2) + 3 \cdot p(3) + 4 \cdot p(4) + 5 \cdot p(5) + 6 \cdot p(6)$$

$$= \sum_{k=1}^{6} k \cdot p(k)$$

Probability: Example

Now, we play a dice game with the following rule for a gain:

$f(1)$ = ₩10,000, $f(2)$ = ₩5,000, $f(3) = f(4) = f(5) = f(6) = 0$

- Expectation for the gain:

$\langle f \rangle =$ ₩10,000×(1/6) + ₩5,000×(1/6) + 0×(4/6) = ₩2,500

$$\langle f \rangle = f(1)p(1) + f(2)p(2) + f(3)p(3)$$
$$+ f(4)p(4) + f(5)p(5) + f(6)p(6)$$
$$= \sum_{k=1}^{6} f(k)p(k)$$

Summary

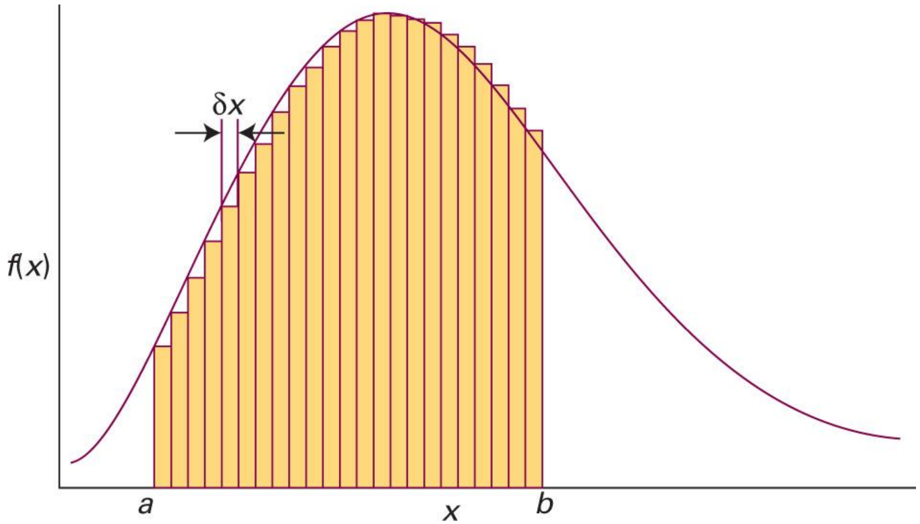
	Discrete
Variable	$k = 1,2, \dots , N$
Probability	$p(k)$
Normalization	$\sum_{k=1}^{N} p(k) = 1$
Average	$\langle k \rangle = \sum_{k=1}^{N} kp(k)$
Expectation	$\langle f \rangle = \sum_{k=1}^{N} f(k)p(k)$

From Discrete to Continuous

	Discrete	Continuous
Variable	$k = 1,2,\ldots,N$	x: real in $[x_1, x_2]$
Probability	$p(k)$	
Normalization	$\sum_{k=1}^{N} p(k) = 1$	
Average	$\langle k \rangle = \sum_{k=1}^{N} kp(k)$	
Expectation	$\langle f \rangle = \sum_{k=1}^{N} f(k)p(k)$	

Integration

"The areas under curves"



$$\lim_{\delta x \to 0} \sum_i f(x_i)\delta x = \int_a^b f(x)\,\mathrm{d}x$$

(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., the chemist's toolkit 4, Sketch 1)

Common Integrals (Part A)

	Indefinite integral*	Constraint	Definite integral
Algebraic functions			
A.1	$\int x^n \mathrm{d}x = \frac{1}{n+1}x^{n+1} + c$	$n \neq -1$	$\int_a^b x^n \,\mathrm{d}x = \frac{1}{n+1}(b^{n+1} - a^{n+1})$
A.2	$\int \frac{1}{x}\mathrm{d}x = \ln x + c$		$\int_a^b \frac{1}{x}\mathrm{d}x = \ln\frac{b}{a}$
A.3	$\int \frac{1}{(A-x)(B-x)}\mathrm{d}x = \frac{1}{B-A}\ln\frac{B-x}{A-x} + c$	$A \neq B$	$\int_a^b \frac{1}{(A-x)(B-x)}\mathrm{d}x = \frac{1}{B-A}\ln\frac{(B-b)(A-a)}{(A-b)(B-a)}$
Exponential functions			
E.1	$\int \mathrm{e}^{-kx}\mathrm{d}x = -\frac{1}{k}\mathrm{e}^{-kx} + c$		$\int_a^b \mathrm{e}^{-kx}\,\mathrm{d}x = \frac{1}{k}(\mathrm{e}^{-ka} - \mathrm{e}^{-kb})$
E.2	$\int x\mathrm{e}^{-kx}\mathrm{d}x = -\frac{1}{k^2}\mathrm{e}^{-kx} - \frac{x}{k}\mathrm{e}^{-kx} + c$		$\int_a^b x\mathrm{e}^{-kx}\,\mathrm{d}x = -\frac{1}{k^2}(\mathrm{e}^{-kb} - \mathrm{e}^{-ka}) - \frac{1}{k}(b\mathrm{e}^{-kb} - a\mathrm{e}^{-ka})$
E.3		$n \geq 0,\ k > 0$	$\int_0^\infty x^n\mathrm{e}^{-kx}\,\mathrm{d}x = \frac{n!}{k^{n+1}}$ $\quad n! = n(n-1)\ldots1;\ 0! = 1$
E.4			$\int_0^\infty \frac{x^4\mathrm{e}^x}{(\mathrm{e}^x - 1)^2}\mathrm{d}x = \frac{4\pi^4}{15}$
Gaussian functions			
G.1			$\int_0^\infty \mathrm{e}^{-kx^2}\,\mathrm{d}x = \frac{1}{2}\left(\frac{\pi}{k}\right)^{1/2}$
G.2	$\int x\mathrm{e}^{-kx^2}\,\mathrm{d}x = -\frac{1}{2k}\mathrm{e}^{-kx^2} + c$		$\int_0^\infty x\mathrm{e}^{-kx^2}\,\mathrm{d}x = \frac{1}{2k}$
G.3			$\int_0^\infty x^2\mathrm{e}^{-kx^2}\,\mathrm{d}x = \frac{1}{4}\left(\frac{\pi}{k^3}\right)^{1/2}$
G.4	$\int x^3\mathrm{e}^{-kx^2}\,\mathrm{d}x = -\frac{1}{2k^2}\mathrm{e}^{-kx^2} - \frac{x^2}{2k}\mathrm{e}^{-kx^2} + c$	$k > 0$	$\int_0^\infty x^3\mathrm{e}^{-kx^2}\,\mathrm{d}x = \frac{1}{2k^2}$
G.5		$k > 0$	$\int_0^\infty x^4\mathrm{e}^{-kx^2}\,\mathrm{d}x = \frac{3}{8k^2}\left(\frac{\pi}{k}\right)^{1/2}$

Probability Distributions

	Discrete	Continuous
Variable	$k = 1,2,\dots,N$	x: real in $[x_1, x_2]$
Probability	$p(k)$	probability density $p(x)\mathrm{d}x$
Normalization	$\sum_{k=1}^{N} p(k) = 1$	$\int_{x_1}^{x_2} p(x)\,\mathrm{d}x = 1$
Average	$\langle k \rangle = \sum_{k=1}^{N} kp(k)$	$\langle x \rangle = \int_{x_1}^{x_2} xp(x)\,\mathrm{d}x$
Expectation	$\langle f \rangle = \sum_{k=1}^{N} f(k)p(k)$	$\langle f \rangle = \int_{x_1}^{x_2} f(x)p(x)\,\mathrm{d}x$

Properties of Probabilities

- $\langle kf(x)\rangle = k\langle f(x)\rangle$

$$\langle kf(x)\rangle = \int_{x_1}^{x_2} [kf(x)]p(x)\,\mathrm{d}x = k\int_{x_1}^{x_2} f(x)p(x)\,\mathrm{d}x = k\langle f(x)\rangle$$

- $\langle f(x) + g(x)\rangle = \langle f(x)\rangle + \langle g(x)\rangle$

$$\langle f(x) + g(x)\rangle = \int_{x_1}^{x_2} [f(x) + g(x)]p(x)\,\mathrm{d}x$$

$$= \int_{x_1}^{x_2} f(x)p(x)\,\mathrm{d}x + \int_{x_1}^{x_2} g(x)p(x)\,\mathrm{d}x = \langle f(x)\rangle + \langle g(x)\rangle$$

Boltzmann Distribution

- Probability density for energy E

$$f(E) \propto e^{-E/k_\mathrm{B}T}$$

$$f(E) = Ke^{-E/k_\mathrm{B}T}$$

- $f(E)\mathrm{d}E$: the probability that a molecule has an energy in the range $[E, E + \mathrm{d}E]$

Probability Density for Velocity

$$f(E) = Ke^{-E/k_\mathrm{B}T}$$

Since the energy of a molecule = its kinetic energy,

$$E = \frac{1}{2}mv^2 = \frac{1}{2}mv_x^2 + \frac{1}{2}mv_y^2 + \frac{1}{2}mv_z^2$$

Probability density for velocity (v_x, v_y, v_z):

$$f(v_x, v_y, v_z) = Ke^{-mv_x^2/2k_\mathrm{B}T}e^{-mv_y^2/2k_\mathrm{B}T}e^{-mv_z^2/2k_\mathrm{B}T}$$

Separation of Variables

$$f\left(v_x, v_y, v_z\right) = Ke^{-mv_x^2/2k_\mathrm{B}T}e^{-mv_y^2/2k_\mathrm{B}T}e^{-mv_z^2/2k_\mathrm{B}T}$$

Let $f\left(v_x, v_y, v_z\right) = f(v_x)f\left(v_y\right)f(v_z)$, where

$$f(v_x) = K_x e^{-mv_x^2/2k_\mathrm{B}T}$$

$$f\left(v_y\right) = K_y e^{-mv_y^2/2k_\mathrm{B}T}$$

$$f(v_z) = K_z e^{-mv_z^2/2k_\mathrm{B}T}$$

such that

$$K = K_x K_y K_z$$

Determination of K's

$$f(v_x) = K_x e^{-mv_x^2/2k_BT}$$

Use the normalization condition.

Since the range of v_x is $(-\infty, \infty)$,

$$\int_{-\infty}^{\infty} f(v_x)\,\mathrm{d}v_x = 1$$

$$\int_{-\infty}^{\infty} K_x e^{-mv_x^2/2k_\mathrm{B}T}\,\mathrm{d}v_x = 1$$

Determination of K's

$$\int_{-\infty}^{\infty} f(v_x)\,\mathrm{d}v_x = 1$$

$$\int_{-\infty}^{\infty} K_x e^{-mv_x^2/2k_\mathrm{B}T}\,\mathrm{d}v_x = 1$$

$$K_x \int_{-\infty}^{\infty} e^{-mv_x^2/2k_\mathrm{B}T}\,\mathrm{d}v_x = 1$$

$$K_x \left(\frac{2\pi k_\mathrm{B}T}{m}\right)^{1/2} = 1$$

$$K_x = \left(\frac{m}{2\pi k_\mathrm{B}T}\right)^{1/2}$$

Normalized Expression

$$K_x = \left(\frac{m}{2\pi k_\mathrm{B}T}\right)^{1/2} = K_y = K_z$$

$$f(v_x, v_y, v_z) = K_x K_y K_z e^{-mv_x^2/2k_\mathrm{B}T} e^{-mv_y^2/2k_\mathrm{B}T} e^{-mv_z^2/2k_\mathrm{B}T}$$

$$= \left(\frac{m}{2\pi k_\mathrm{B}T}\right)^{3/2} e^{-mv_x^2/2k_\mathrm{B}T} e^{-mv_y^2/2k_\mathrm{B}T} e^{-mv_z^2/2k_\mathrm{B}T}$$

$$= \left(\frac{m}{2\pi k_\mathrm{B}T}\right)^{3/2} e^{-m(v_x^2+v_y^2+v_z^2)/2k_\mathrm{B}T}$$

Velocity to Speed

- $f(v_x, v_y, v_z)\mathrm{d}v_x\mathrm{d}v_y\mathrm{d}v_z$:

  Probability that a molecule has a velocity in the range

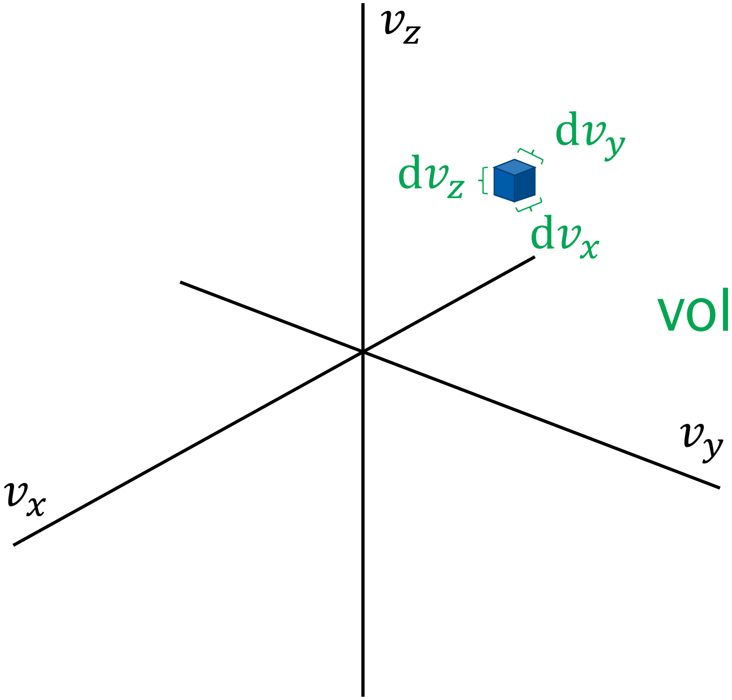
  $$[v_x, v_x + \mathrm{d}v_x], [v_y, v_y + \mathrm{d}v_y], [v_z, v_z + \mathrm{d}v_z]$$

- $f(v)\mathrm{d}v$:

  Probability that a molecule has a speed in the range $[v, v + \mathrm{d}v]$

Can we obtain $f(v)\mathrm{d}v$ from $f(v_x, v_y, v_z)\mathrm{d}v_x\mathrm{d}v_y\mathrm{d}v_z$?

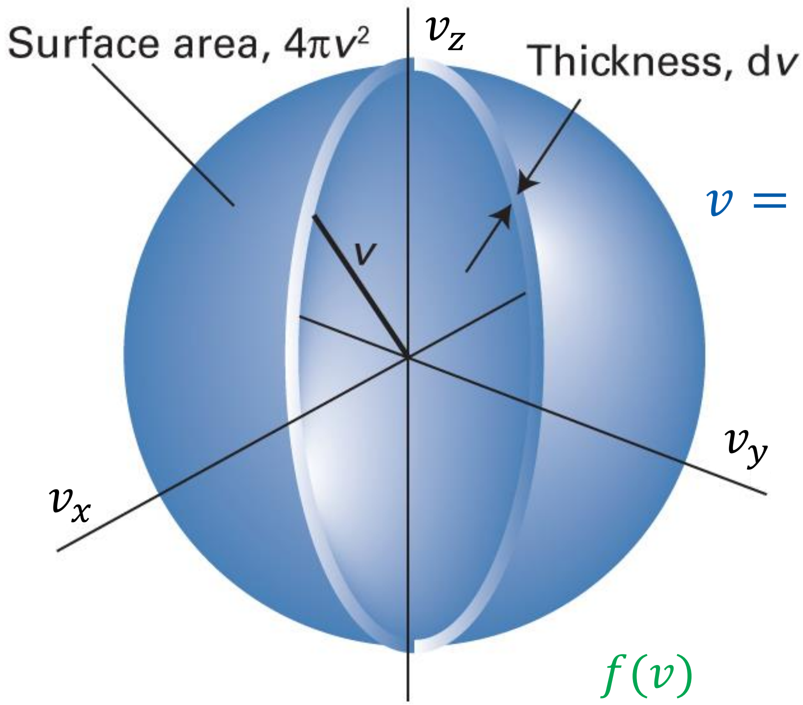
Velocity Space



volume: $\mathrm{d}v_x\mathrm{d}v_y\mathrm{d}v_z$

$f(v_x, v_y, v_z)\,\mathrm{d}v_x\mathrm{d}v_y\mathrm{d}v_z$ = function × volume

Velocity Space



$$v = \sqrt{v_x^2 + v_y^2 + v_z^2}$$

$$\text{function} \times \text{volume} = \overbrace{f\left(v_x, v_y, v_z\right) \times 4\pi v^2}^{f(v)}\mathrm{d}v$$

(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., Figure 1B.3)

Maxwell-Boltzmann Distribution

$$f(v) = f(v_x, v_y, v_z) 4\pi v^2$$

$$f(v) = 4\pi \left(\frac{m}{2\pi k_{\mathrm{B}} T}\right)^{3/2} v^2 e^{-m(v_x^2+v_y^2+v_z^2)/2k_{\mathrm{B}}T}$$

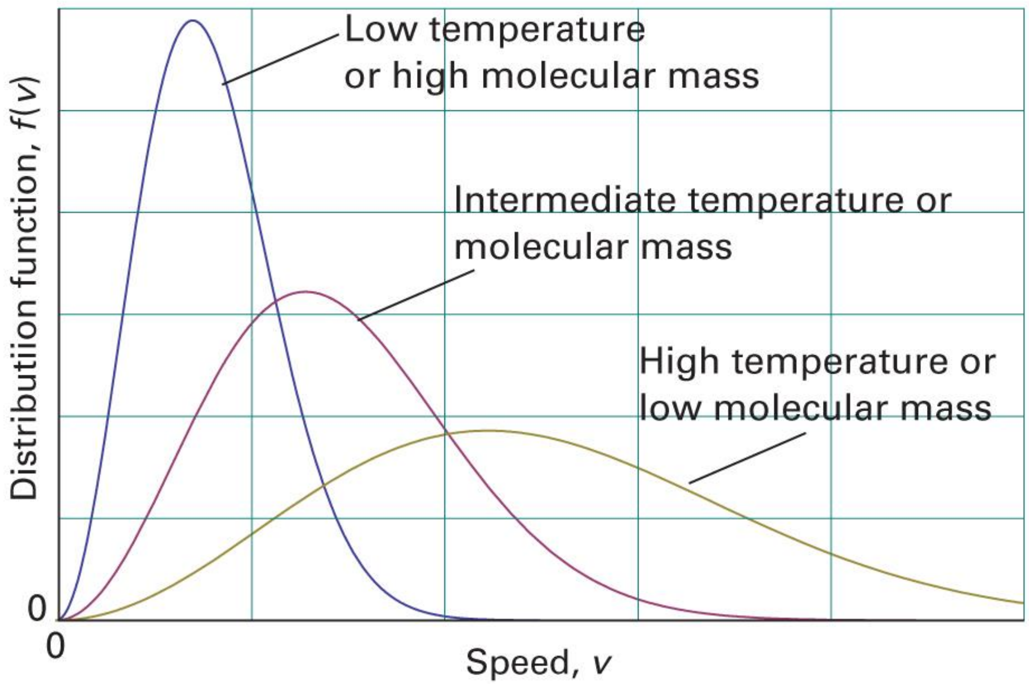
$$f(v) = 4\pi \left(\frac{m}{2\pi k_{\mathrm{B}} T}\right)^{3/2} v^2 e^{-mv^2/2k_{\mathrm{B}}T}$$

$R = k_{\mathrm{B}} N_{\mathrm{A}}$
$M = m N_{\mathrm{A}}$

$$f(v) = 4\pi \left(\frac{M}{2\pi R T}\right)^{3/2} v^2 e^{-Mv^2/2RT}$$

Interpretations

$$f(v) = 4\pi\left(\frac{M}{2\pi RT}\right)^{3/2} v^2 e^{-Mv^2/2RT}$$



(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., Figure 1B.4)

Characteristic Speeds

1. Mean speed $\langle v \rangle = v_{\mathrm{mean}}$

$$\langle v \rangle = \int_0^\infty v f(v)\,\mathrm{d}v$$

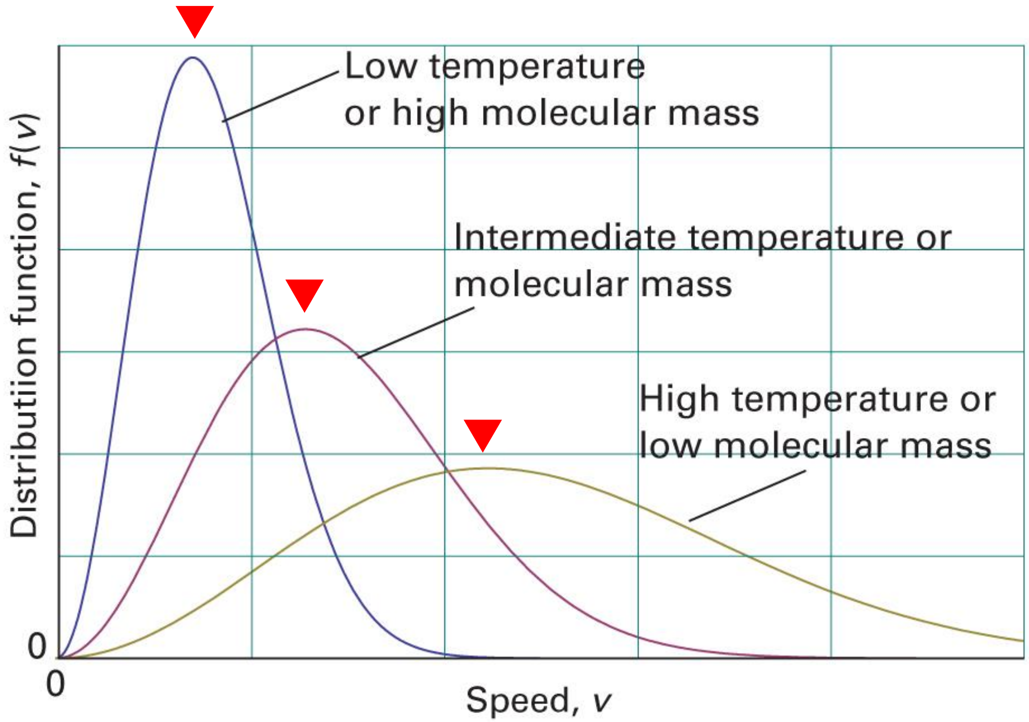
2. Root-mean-square speed v_{rms}

$$v_{\mathrm{rms}} = \sqrt{\langle v^2 \rangle}$$

$$\langle v^2 \rangle = \int_0^\infty v^2 f(v)\,\mathrm{d}v$$

Characteristic Speeds

3. Most probable speed v_{mp}



$$\frac{\mathrm{d}f(v)}{\mathrm{d}v} = 0$$

(Image: Atkins *et al.*, ***Atkins' Physical Chemistry***, 11th ed., Figure 1B.4)

Characteristic Speeds: KMT

1. Mean speed

$$\langle v \rangle = \left(\frac{8RT}{\pi M}\right)^{1/2}$$

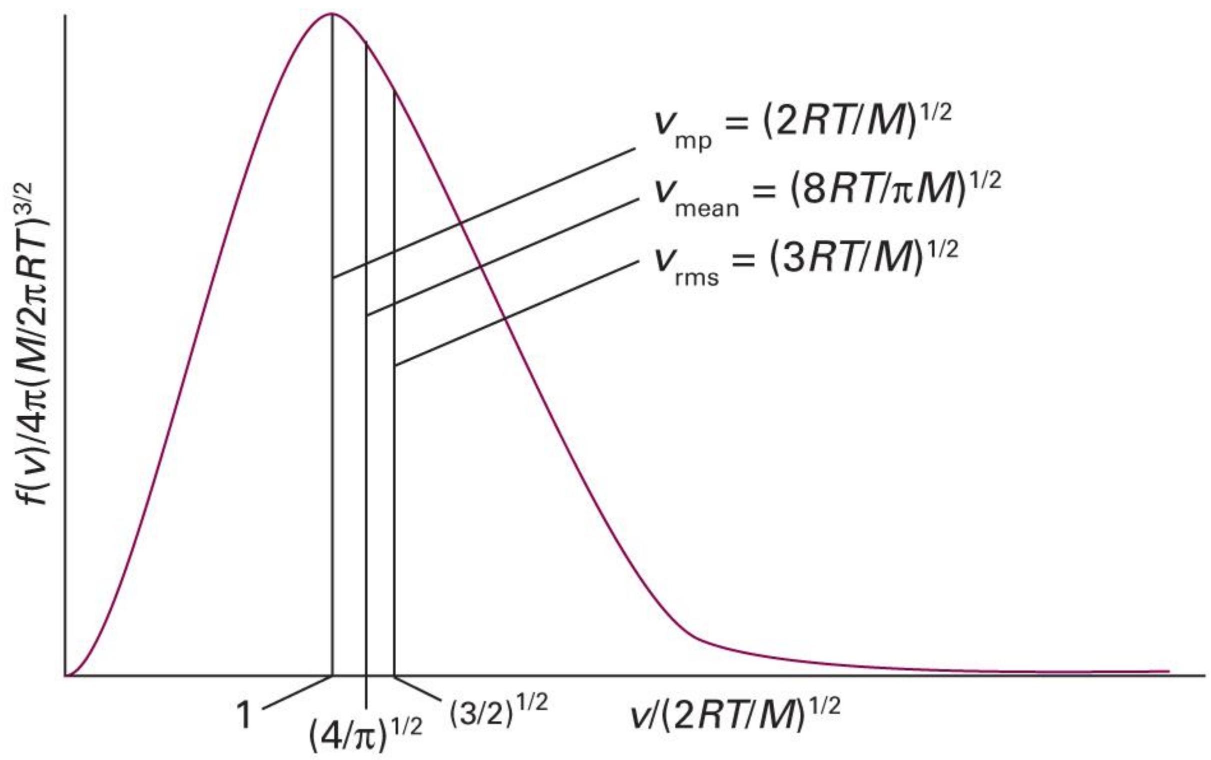
2. Root-mean-square speed

$$v_{\mathrm{rms}} = \left(\frac{3RT}{M}\right)^{1/2}$$

3. Most probable speed

$$v_{\mathrm{mp}} = \left(\frac{2RT}{M}\right)^{1/2}$$

Characteristic Speeds: KMT



(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., Figure 1B.6)

Example 1B.1

Calculate v_{rms} and the mean speed, $\langle v \rangle$, of N_2 molecules at 25 °C.

For N_2, M = 28.02 g mol^{-1} = 0.02802 kg mol^{-1}

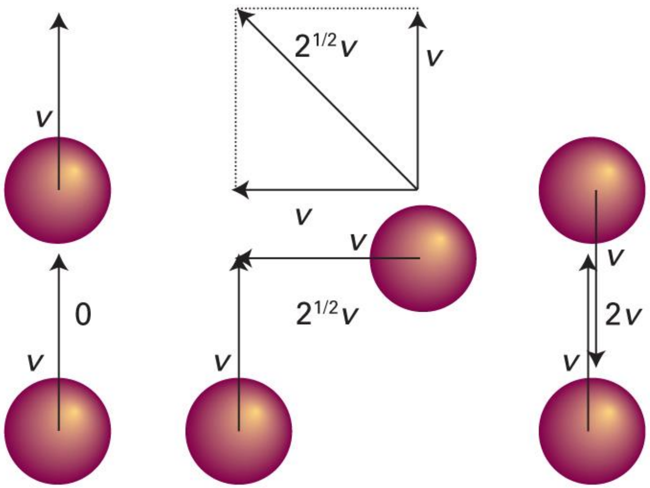
T = (273 + 25) K = 298 K

$$v_{\text{rms}} = \left(\frac{3RT}{M}\right)^{1/2} = \left\{\frac{3 \times (8.3145\ \text{J K}^{-1}\ \text{mol}^{-1}) \times (298\ \text{K})}{0.02802\ \text{kg mol}^{-1}}\right\}^{1/2} = 515\ \text{m s}^{-1}$$

$$\langle v \rangle = \left(\frac{8RT}{\pi M}\right)^{1/2} = \left\{\frac{8 \times (8.3145\ \text{J K}^{-1}\ \text{mol}^{-1}) \times (298\ \text{K})}{3.1416 \times 0.02802\ \text{kg mol}^{-1}}\right\}^{1/2} = 475\ \text{m s}^{-1}$$

Mean Relative Speed

“Mean speed with which a molecule approaches another”



$$v_{\mathrm{rel}} = 2^{1/2}\langle v\rangle = \left(\frac{16RT}{\pi M}\right)^{1/2}$$

(Image: Atkins *et al.*, ***Atkins’ Physical Chemistry***, 11th ed., Figure 1B.7)

Mean Relative Speed

For two molecules of different types (A and B),

$$v_{\mathrm{rel}} = \left(\frac{8k_{\mathrm{B}}T}{\pi\mu}\right)^{1/2}$$

where reduced mass $\mu = m_{\mathrm{A}}m_{\mathrm{B}}/(m_{\mathrm{A}} + m_{\mathrm{B}})$

If $m_{\mathrm{A}} = m_{\mathrm{B}}$,

$$\mu = m_{\mathrm{A}}^2/(2m_{\mathrm{A}}) = m_{\mathrm{A}}/2$$

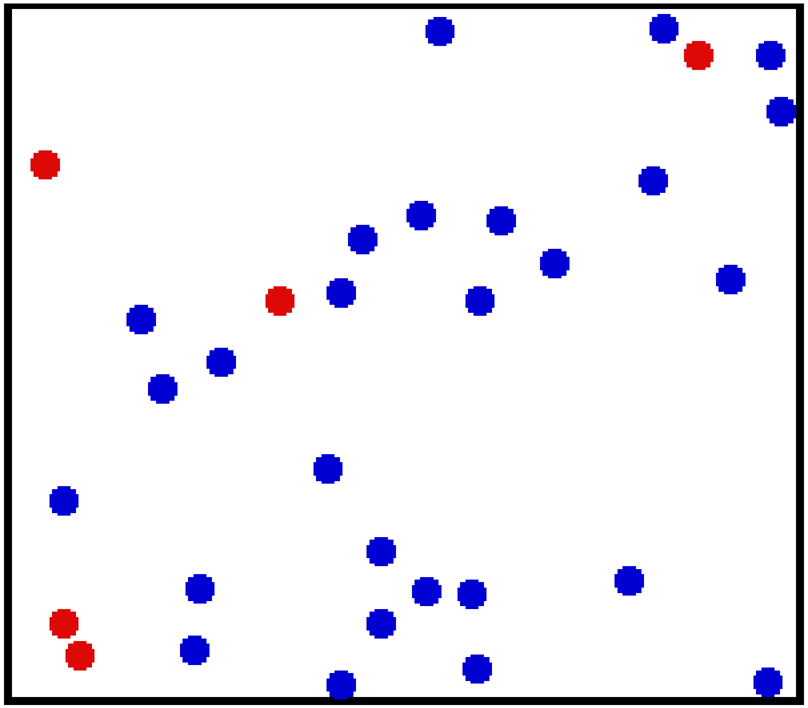
$$v_{\mathrm{rel}} = \left(\frac{16k_{\mathrm{B}}T}{\pi m}\right)^{1/2} = \left(\frac{16RT}{\pi M}\right)^{1/2}$$

Brief Illustration 1B.1

The mean speed of N_2 molecules at 25 °C is 475 m s^{-1}. What is their relative mean speed?

$$v_{\mathrm{rel}} = 2^{1/2}\langle v \rangle = \sqrt{2} \times (475\ \mathrm{m\ s^{-1}}) = 671\ \mathrm{m\ s^{-1}}$$

Application: Collisions



(Image: https://commons.wikimedia.org/wiki/File:Translational_motion.gif)

Consequences of Collisions

- Molecule-molecule collisions

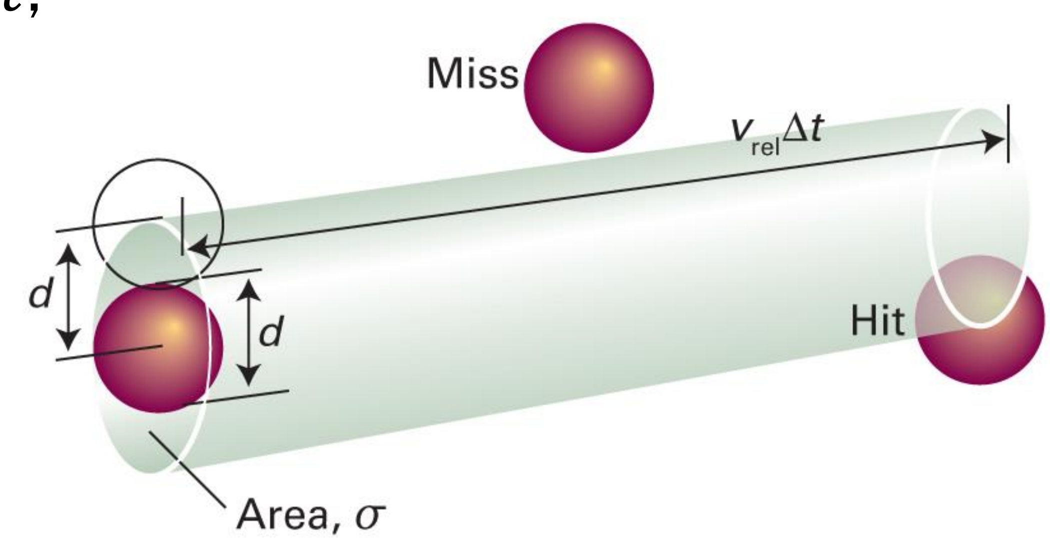
Molecular origin of chemical reactions

- Molecule-wall collisions

Molecular origin of pressure

Molecule-Molecule Collisions

For time Δt,

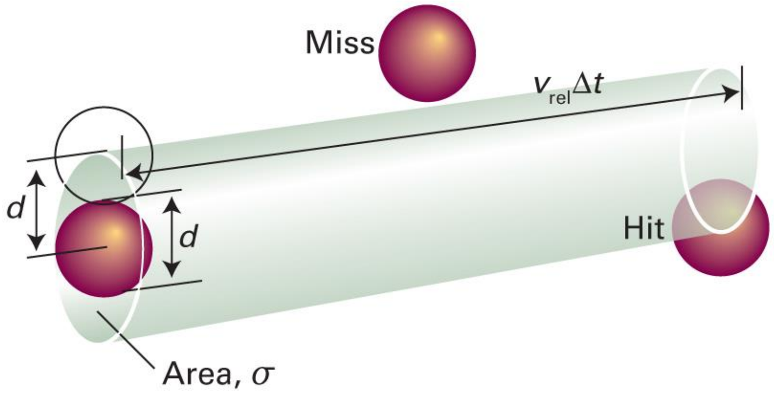


d = collision diameter

$\sigma = \pi d^2$ = cross-sectional area

(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., Figure 1B.8)

Collision Frequency



- Volume of the collision tube = $\sigma \times v_{\text{rel}}\Delta t$
- Number of molecules in the tube = volume × density

  = $(\sigma v_{\text{rel}}\Delta t) \times (nN_{\text{A}}/V)$
- Collision frequency $z = \sigma v_{\text{rel}} nN_{\text{A}}/V$

T- and p-dependences of z

$$z = \frac{\sigma v_{\text{rel}} n N_{\text{A}}}{V} = \sigma v_{\text{rel}} N_{\text{A}} \frac{n}{V}$$

Since $v_{\text{rel}} = (16RT/\pi M)^{1/2}$,

$$z = \sigma N_{\text{A}} \frac{n}{V} \left(\frac{16RT}{\pi M} \right)^{1/2} \propto \sqrt{T} \text{ (at constant } V\text{)}$$

Since $pV = nRT$,

$$z = \sigma v_{\text{rel}} N_{\text{A}} \frac{p}{RT} = \sigma v_{\text{rel}} \frac{p}{k_{\text{B}} T} \propto p \text{ (at constant } T\text{)}$$

Brief Illustration 1B.2

What is the collision frequency of an N_2 molecule in a sample at 1.00 atm and 25 °C? (σ = 0.45 nm^2)

From Brief Illustration 1B.1, v_{rel} = 671 m s^{-1}.

$$z = \frac{\sigma v_{\mathrm{rel}} p}{k_{\mathrm{B}} T} = \frac{(0.45 \times 10^{-18}\ \mathrm{m}^2) \times (671\ \mathrm{m\ s}^{-1}) \times (1.01 \times 10^5\ \mathrm{Pa})}{(1.381 \times 10^{-23}\ \mathrm{J\ K}^{-1}) \times (298\ \mathrm{K})}$$

$$z = 7.4 \times 10^9\ \mathrm{s}^{-1}$$

Mean Free Path, λ

"Average distance a molecule travels between collisions"

$$\lambda = \text{speed} \times \text{time} = v_{\mathrm{rel}} \times (1/z) = v_{\mathrm{rel}}/z$$

Hence,

$$\lambda = \frac{k_B T}{\sigma p}$$

Brief Illustration 1B.3

What is the mean free path of an N_2 molecule in a sample at 1.00 atm and 25 °C?

From Brief Illustration 1B.1, v_{rel} = 671 m s^{-1}.

From Brief Illustration 1B.2, z = 7.4 × 10^{9} s^{-1}.

$$\lambda = \frac{v_{\mathrm{rel}}}{z} = \frac{671\ \mathrm{m\ s^{-1}}}{7.4 \times 10^{9}\ \mathrm{s^{-1}}} = 9.1 \times 10^{-8}\ \mathrm{m}$$

Prelude to Chemical Kinetics

$$z = \sigma v_{\mathrm{rel}} N_{\mathrm{A}} \frac{n}{V}$$

Suppose that we have two types of molecules, A and B.

The collision rate of one A molecule with B molecules is

$$z = \sigma v_{\mathrm{rel}} N_{\mathrm{A}} \frac{n_{\mathrm{B}}}{V}$$

Hence, the collision density $Z_{\mathrm{AB}} = z(N_{\mathrm{A}} n_{\mathrm{A}}/V)$

$$Z_{\mathrm{AB}} = \sigma v_{\mathrm{rel}} N_{\mathrm{A}}^2 \frac{n_{\mathrm{A}}}{V} \frac{n_{\mathrm{B}}}{V} = \sigma \left(\frac{8k_B T}{\pi\mu}\right)^{1/2} N_{\mathrm{A}}^2 [\mathrm{A}][\mathrm{B}] \propto [\mathrm{A}][\mathrm{B}]$$